

Certificate of Analysis

PRODUCT DETAILS

Product Name	Exosomes (RejuvWELL*)
Batch	
Manufacturing Date	
Expiry Date	
Dosage Form	Clear-colourless solution
Storage Temperature	4 – 8°C

TEST	SPECIFICATION	TEST RESULTS	REMARKS
Infectious Diseases Screening	HBsAg; HCV Ab; HIV1/2 Ag/Ab; CMV Ab; RPR	Not Detected / Non-Reactive	QUALITY CONTROL APPROVED
Sterility	No cloudiness of the test media	Negative	QUALITY CONTROL APPROVED
Bacterial Endotoxin Test (LAL)	Endotoxin sensitivity 0.125 EU/mL	Negative	QUALITY CONTROL APPROVED
Mycoplasma	Mean of absorbance <0.05	Negative	QUALITY CONTROL APPROVED
Immunophenotyping	Positive marker – CD9, CD63, CD81	Detected	QUALITY CONTROL APPROVED
Nanoparticle Tracking Analysis (NTA)	Exosome particle size range 30nm – 200nm	10 x 10 ⁹ particles/mL	QUALITY CONTROL APPROVED

The cell product delivered by Malaysian Genomics Resource Centre Berhad is manufactured and tested for research purpose only. Product is processed in a facility operating under current Good Manufacturing Practice (cGMP) guidelines in accordance with PIC/S standards.

Storage recommendation:

- 1) Refrigerate the vials in 4-8°C at all times upon received
- 2) Keep away the vials from direct heat or sunlight
- 3) Avoid prolong exposure in room condition
- 4) It is recommended to place the vials in freezer for prolonged storage

*General Exosomes fortified with Tretinoin & Melatonin

Authorised Signature:

**Quality Control Department
Date:**

Certificate of Analysis

Supporting Information

STERILITY TEST ^{6, 8}

Aerobic microorganisms are incubated in Soybean Casein Digest Medium at 20°C - 25°C for 14 days.
Facultative microorganisms are incubated in Fluid Thioglycollate Medium at 30°C - 35°C for 14 days.

BACTERIAL ENDOTOXIN TEST (LIMULUS AMOEBOCYTE LYSATE, LAL) TEST ^{3,6}

Negative findings for Gram-negative bacteria in cell culture based on FDA Reference of Standard Endotoxin Guideline.

MYCOPLASMA TEST ^{4,6}

Negative findings for mycoplasma species *M. hyorhinis*, *M. arginini*, *M. fermentans*, *M. orale*, *M. Pirum*, *M. hominis*, *M. salivarium* and *A. laidlawii*.

IMMUNOPHENOTYPING

Marker	Positive		
	CD9+	CD63+	CD81+
Result	+	+	+

‘+’ indicates the presence of the marker in the cell.

NANOPARTICLE TRACKING ANALYSIS (NTA) ^{7,8}

Established method for the direct and real-time visualization and analysis of the nanoparticles size distribution and concentration in liquids.



REFERENCES

- Chettimada, Sukrutha, Lorenz, David R., Misra, Vikas, Dillon, Simon T., Reeves, R. Keith, Manickam, Cordelia, 2018, Exosome markers associated with immune activation and oxidative stress in HIV patients on antiretroviral therapy.
- Joana Maia, Silvia Batista, Nuno Couto, Ana C. Gregório, Cristian Bodo, Julia Etzanowska, Maria Carolina Strano Moraes, Bruno Costa-Silva, 2020, Employing Flow Cytometry to Extracellular Vesicles Sample Microvolume Analysis and Quality Control.
- Bacterial Endotoxins. European Pharmacopoeia 5.0; <https://gmp.eu/validation/Method/LAL/EUPHARMACPOE05a.pdf>
- US Pharmacopeia <83> Mycoplasma Tests
- ICH Harmonised Tripartite Guideline. Specifications: Test Procedures and Acceptance Criteria for Biotechnological/ Biological Products Q6B.
- US Pharmacopeia <71> Sterility Test
- Neha Chopra, Ibrahim Dutt Arora, Namrata Jain, Poonam Yadav, Salma Wajid, Surinder P. Singh, Sangeta Choudhary, 2019, Biophysical Characterization and Drug Delivery Potential of Exosomes from Human Wharton's Jelly-Derived Mesenchymal Stem Cells.
- Dae Hyun Ha, Hyun-keun Kim, Joon Lee, Hyuck-Hoon Kwon, Gyeong-Hun Park, Seon-Ho Seong Yang, Jee Yoon Jung, Hyoung Choi, Jun Ho Lee, Sumi Sung, Yong Weon Yi, Byung Seung Cho, 2020, Mesenchymal Stem/Stromal Cell-Derived Exosomes for Immunomodulatory Therapeutics and Skin Regeneration.